

MIT OpenCourseWare

5.62

Physical Chemistry II 5.62 — Physical Chemistry II May 2026 **EXTREMELY****HARD – MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (5 points) Prove the fundamental theorem concerning Reaction Dynamics in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
2. (5 points) Analyze the limitations of standard approaches to Molecular Beams when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (4 points) Analyze the limitations of standard approaches to Laser Chemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
4. (3 points) Prove the fundamental theorem concerning Femtochemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (4 points) Analyze the limitations of standard approaches to Surface Chemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (5 points) Analyze the limitations of standard approaches to Reaction Dynamics when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (8 points) Analyze the limitations of standard approaches to Molecular Beams when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
8. (2 points) Prove the fundamental theorem concerning Laser Chemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

9. (4 points) Prove the fundamental theorem concerning Femtochemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
10. (4 points) Construct an original proof-level problem involving Surface Chemistry for 5.62 (Physical Chemistry II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
11. (2 points) Prove the fundamental theorem concerning Reaction Dynamics in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
12. (4 points) Construct an original proof-level problem involving Molecular Beams for 5.62 (Physical Chemistry II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
13. (5 points) Analyze the limitations of standard approaches to Laser Chemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
14. (2 points) Analyze the limitations of standard approaches to Femtochemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
15. (2 points) Construct an original proof-level problem involving Surface Chemistry for 5.62 (Physical Chemistry II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (3 points) Prove the fundamental theorem concerning Reaction Dynamics in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
17. (5 points) Analyze the limitations of standard approaches to Molecular Beams when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
18. (4 points) Prove the fundamental theorem concerning Laser Chemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (4 points) Analyze the limitations of standard approaches to Femtochemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.

20. (5 points) Analyze the limitations of standard approaches to Surface Chemistry when applied to edge cases in 5.62. Construct explicit counterexamples showing where intuitive reasoning fails.
21. (7 points) Construct an original proof-level problem involving Reaction Dynamics for 5.62 (Physical Chemistry II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (3 points) Prove the fundamental theorem concerning Molecular Beams in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
23. (2 points) Prove the fundamental theorem concerning Laser Chemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
24. (5 points) Prove the fundamental theorem concerning Femtochemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
25. (3 points) Prove the fundamental theorem concerning Surface Chemistry in the context of 5.62 (Physical Chemistry II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.

Solutions

Solution 1:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Beams. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Laser Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 4:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Surface Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Reaction Dynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Beams. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 8:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 9:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 10:

Solution approach: First recall the definitions and key theorems related to Surface Chemistry from 5.62. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 11:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 12:

Solution approach: First recall the definitions and key theorems related to Molecular Beams from 5.62. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 13:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Laser Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 14:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Femtochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 15:

Solution approach: First recall the definitions and key theorems related to Surface Chemistry from 5.62. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

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The edge case analysis reveals the subtle assumptions underlying standard treatments of Molecular Beams. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 18:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

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The edge case analysis reveals the subtle assumptions underlying standard treatments of Femtochemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 20:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Surface Chemistry. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Reaction Dynamics from 5.62. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

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The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 24:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 25:

The proof follows from the definitions and properties established in 5.62. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.